№ 4(58) 2026
НАУКА і ТЕХНІКА СЬОГОДНІ
ISSN 2786-6025 Online
UDC 004.8
https://doi.org/10.52058/2786-6025-2026-4(58)-2670-2682

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# CYBERSECURITY RISKS OF GENERATIVE AI AND AGENTIC SYSTEMS: A SYSTEMATIC ANALYSIS BASED ON OWASP FRAMEWORK

**Abstract.** The rapid adoption of generative artificial intelligence and autonomous agentic systems has fundamentally transformed the cybersecurity landscape, exposing a widening gap between the pace of technological integration and the maturity of corresponding control mechanisms. Empirical trends indicate that a substantial share of enterprise AI usage occurs through "shadow AI" channels—particularly via personal accounts—resulting in significant data leakage, predominantly through manual copy–paste practices. Beyond human-driven risks, the transition toward autonomous agents introduces systemic vulnerabilities, including indirect prompt injection, unauthorized goal modification, and the abuse of delegated privileges, as formalized in the OWASP Top 10 for Agentic Applications (2026). These risks are further exacerbated by weaknesses in the AI supply chain, retrieval-augmented generation (RAG) context poisoning, and over-privileged OAuth integrations, as evidenced by recent high-profile breaches involving third-party AI services. In response, the cybersecurity paradigm is shifting from traditional Data Loss Prevention (DLP) approaches toward multi-layered defensive architectures grounded in Zero Trust principles, policy-as-code enforcement, and sandboxed execution. Concurrently, leading organizations are institutionalizing "trusted access" frameworks and resource-oriented security models, wherein system resilience is evaluated through an economic lens that compares the computational and operational cost of attacks against defensive investments. Ultimately, securing agentic AI ecosystems requires a holistic rethinking of security architecture, integrating deterministic runtime controls with continuous observability, provenance tracking, and adaptive policy enforcement. The growing convergence of large language models with external tools, APIs, and persistent memory further dissolves traditional system boundaries, reconfiguring AI agents into deeply interconnected socio-technical actors. This transformation challenges conventional perimeter-based security models, necessitating dynamic trust recalibration across all layers of interaction, including data retrieval, reasoning, and action execution.

ISSN 2786-6025 Online

**Keywords:** Agentic Systems; Generative Artificial Intelligence; Large Language Models (LLMs); Data Leakage; Prompt Injection; OWASP; Retrieval-Augmented Generation (RAG); Context Poisoning; Supply Chain Security; AI Agents; Multi-Agent Systems; Zero Trust Architecture.

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# КІБЕРБЕЗПЕКОВІ РИЗИКИ ГЕНЕРАТИВНОГО ШТУЧНОГО ІНТЕЛЕКТУ ТА АГЕНТНИХ СИСТЕМ: СИСТЕМАТИЧНИЙ АНАЛІЗ НА ОСНОВІ ФРЕЙМВОРКУ OWASP

**Анотація.** Стрімке впровадження генеративного штучного інтелекту та автономних агентних систем докорінно трансформувало ландшафт кібербезпеки, виявивши зростаючий розрив між темпами технологічної інтеграції та зрілістю відповідних механізмів контролю. Емпіричні тенденції свідчать, що значна частка використання ШІ в корпоративному середовищі відбувається через канали «shadow AI» — зокрема через особисті облікові записи — що призводить до суттєвих витоків даних, переважно внаслідок ручних операцій копіювання та вставлення. Окрім ризиків, зумовлених людським фактором, перехід до автономних агентів створює системні вразливості, зокрема непрямі ін'єкції підказок (prompt injection), несанкціоновану модифікацію цілей та зловживання делегованими привілеями, що формалізовано в OWASP Top 10 for Agentic Applications (2026). Ці ризики додатково посилюються слабкими місцями в ланцюгах постачання ШІ, отруєнням контексту в системах RAG та надмірно привілейованими інтеграціями OAuth, що підтверджується нещодавніми резонансними інцидентами компрометації за участю сторонніх AI-сервісів. У відповідь на ці виклики парадигма кібербезпеки зміщується від традиційних підходів Data Loss Prevention (DLP) до багаторівневих захисних архітектур, заснованих на принципах Zero Trust, застосуванні policy-as-code та ізольованому виконанні. Паралельно провідні організації інституціоналізують підходи «trusted access» і ресурсно-орієнтовані моделі безпеки, в яких стійкість систем оцінюється крізь економічну призму — шляхом порівняння обчислювальних і операційних витрат на здійснення атак із інвестиціями в захист. Зрештою, забезпечення безпеки агентних екосистем ШІ потребує комплексного переосмислення архітектури безпеки з інтеграцією детермінованих механізмів контролю під час виконання, безперервної спостережуваності, відстеження походження даних та адаптивного застосування політик. Подальша конвергенція великих мовних моделей із зовнішніми інструментами,

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API та довготривалою пам'яттю додатково розмиває традиційні межі систем, перетворюючи AI-агентів на глибоко інтегровані соціотехнічні сутності. Така трансформація ставить під сумнів ефективність класичних периметрових моделей безпеки та вимагає динамічного переоцінювання довіри на всіх рівнях взаємодії, включно з отриманням даних, міркуванням і виконанням дій.

**Ключові слова:** агентні системи; генеративний штучний інтелект; великі мовні моделі (LLMs); витік даних; ін'єкції підказок; OWASP; retrieval-augmented generation (RAG); отруєння контексту; безпека ланцюга постачання; AI-агенти; мультиагентні системи; архітектура Zero Trust.

**Problem statement.** The rapid integration of generative AI into corporate workflows has created a new class of cybersecurity problems that is not adequately covered by traditional perimeter-based defenses and conventional DLP mechanisms. In practice, employees increasingly interact with AI systems through personal accounts and uncontrolled channels, which significantly raises the risk of confidential data leakage, unauthorized transfer of sensitive information, and misuse of corporate knowledge.

At the same time, the shift from simple chatbots to autonomous agentic systems expands the attack surface by introducing tool access, long-term memory, inter-agent communication, and external service integration, all of which can be manipulated through prompt injection, context poisoning, identity abuse, or supply chain compromise.

As a result, organizations face a dual security challenge: they must protect both human-AI interaction channels and autonomous AI ecosystems that can execute actions with limited supervision. Existing security approaches are often fragmented and do not fully account for the architectural specificity of agentic systems, including dynamic decision-making, dependency on external data sources, and cascading failures across multiple agents. Therefore, there is a need for a systematic analysis of the main threat categories affecting modern AI-driven environments and for the development of comprehensive protection strategies that combine access control, trust verification, isolation, monitoring, and secure handling of retrieved or generated content.

**Analysis of recent studies and publications.** Recent research shows a clear evolution in AI security from isolated prompt-level attacks toward system-level vulnerabilities in LLM-integrated applications. Greshake et al. [1] demonstrated that indirect prompt injection can exploit the blurred boundary between data and instructions in real-world systems, enabling data theft, manipulation of application behavior, worm-like propagation, and other security abuses; their experiments showed practical impact on systems such as Bing Chat and code-completion environments, while also highlighting the lack of effective mitigations.

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A second major direction concerns retrieval-augmented generation (RAG). Zou et al. [2] showed that the knowledge database itself becomes a practical attack surface: by injecting a small number of malicious texts, an attacker can steer the model toward an attacker-chosen answer for a target question. Their results are especially important because they report high attack success even in large databases, and they also show that common defenses such as paraphrasing and perplexity-based detection are insufficient.

More recent work extends this logic to multi-agent architectures. Triedman et al. [3] showed that adversarial content can hijack control and communication inside multi-agent systems, trigger unsafe functionality, and lead to arbitrary code execution or exfiltration of sensitive data; importantly, these attacks can succeed even when individual agents are not directly vulnerable to prompt injection and refuse harmful actions. This indicates that security failures emerge not only from the model itself, but also from orchestration, delegation, and inter-agent control flow.

At the conceptual level, Kato et al. [4] proposed the AI Security Map, which organizes AI security around both information-system aspects and external stakeholder impacts. The paper connects technical compromise with broader harms such as privacy loss, rights violations, and misuse-driven damage, thereby shifting the discussion from model robustness alone to a more holistic view of risk. Li et al. [5] further consolidate this trend by arguing that frontier AI agents weaken traditional assumptions about code-data separation, authority boundaries, and execution predictability. Their analysis emphasizes indirect prompt injection, confused-deputy behavior, and cascading failures in long-running workflows, while recommending layered defenses, sandboxing, deterministic policy enforcement, and new benchmarks for delegation and privilege control. This aligns with broader survey work calling for fine-grained risk assessment and stronger pre-deployment testing for hybrid AI systems.

**The purpose of the article.** The purpose of this article is to analyze the evolving cybersecurity risks associated with the widespread adoption of generative AI and agentic systems in modern organizational environments. It aims to systematize key threat categories—including data leakage through LLM interactions, vulnerabilities in autonomous multi-agent architectures, supply chain compromises, and RAG-specific attack vectors—and to demonstrate how these emerging risks extend beyond traditional security models. Based on recent studies and industry frameworks such as OWASP for agentic applications, the article further seeks to highlight the need for new, comprehensive security approaches tailored to the dynamic and autonomous nature of AI-driven ecosystems.

**Outline of the main material.** The rapid adoption of generative artificial intelligence in corporate workflows during 2024–2026 has significantly changed the cybersecurity landscape. AI-assisted tools have become an increasingly important

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channel for potential data leakage and unauthorized information transfer. According to industry report [6], the adoption of AI technologies in 2025 is outpacing the development of control mechanisms: up to 45% of employees in large organizations utilize neural network models, with over 67% of these interactions occurring through personal accounts (shadow AI) beyond the corporate security perimeter. A key risk factor is the nature of user interaction, with approximately 77% of confidential information transfers occurring through direct copy–paste operations.

Alongside the evolution of AI chatbots, there is a clear shift toward autonomous agent-based systems. A representative example is the open-source framework OpenClaw [7]. This system achieves a high degree of autonomy through an extensive library of community-developed "skills," which enables integration with a wide range of software systems—ranging from graphical editors to banking applications. The advancement of multi-agent systems has further catalyzed the creation of specialized environments for AI-agent interaction, such as the experimental social network Moltbook [8], designed specifically for researching the collective behavior of autonomous entities. However, increased functionality and autonomy introduce new attack vectors that extend beyond traditional security models. The interaction of agents within isolated environments and their access to critical APIs create significant risks of unauthorized action execution and privilege escalation. Consequently, the OWASP Top 10 for Agentic Applications 2026 [9] framework has emerged as the industry benchmark for identifying and analyzing risks within such modern agentic architectures. This creates a dual challenge for modern information security systems: the necessity to neutralize threats stemming from the human factor (leaks via LLMs) and the imperative to secure autonomous multi-agent environments. Addressing these issues requires a transition from classic DLP solutions toward the development of comprehensive security policies that account for the unique characteristics of agentic systems and the dynamic nature of interactions within AI ecosystems.

The OWASP classification for agentic applications identifies a series of typical vulnerabilities related not only to implementation errors but also to the underlying architecture of autonomous decision-making. One of the key risks involves the unauthorized modification of an agent's objectives due to external or internal factors. In such instances, the decision-making process becomes compromised, causing the system to execute actions that contradict its original intent. Such attacks are executed via indirect prompt injection mechanisms, context manipulation, or instruction substitution [1], potentially resulting in security policy violations, data leaks, and uncontrolled behavior.

Another significant category comprises threats related to the improper or malicious use of tools accessible to the agent. Here, the risk lies not in the alteration of goals, but in the potential to coerce the agent into performing hazardous operations

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through APIs, external services, or modules that exceed their intended scope. The consequences include compromised system integrity, unauthorized data modification, or interactions with critical resources. A distinct risk arises from the abuse of the agent's identity and privileges; acting as a separate access entity capable of storing user or service credentials, the agent creates a supplementary vector for bypassing traditional security controls.

A significant class of risks in agentic systems is associated with supply chain vulnerabilities. Such systems rely on large language models, model weights, external tools, datasets, and auxiliary services (e.g., MCP and A2A servers), making their behavior dependent on numerous external components whose trustworthiness is often insufficiently verified. The compromise of any element in this chain may lead to the injection of hidden instructions, malicious code, or unsafe behaviors executed on behalf of the agent.

A representative example of this type of attack—amplified by flaws in the access control model (over-privileged OAuth) and improper data classification (mislabeling sensitive variables as “non-sensitive”)—is a recent large-scale breach at the Vercel cloud platform in April 2026 [10]. The incident originated from the integration of an external AI-driven corporate agent, Context.ai, into a workspace via OAuth authorization. After compromising the third-party service, attackers exploited excessive permissions to infiltrate internal infrastructure. The breach revealed a critical architectural weakness in the handling of environment variables: values assumed to be non-sensitive were neither encrypted nor adequately protected. As a result, attackers were able to enumerate these variables and extract high-value assets, including NPM tokens, private GitHub repository keys, and cryptographic signing keys. This case illustrates the systemic risks inherent in AI supply chains, where autonomous integrations can become vectors for privilege escalation and broader infrastructure compromise.

Risks of unintended code execution constitute a distinct category. They arise when an agent dynamically generates or executes code based on external data or service responses without sufficient environment isolation. In such scenarios, instruction injection and unauthorized access to system resources outside the defined context become possible, including access to the file system or external network resources. This category also includes memory and context attacks, in which an adversary poisons the agent’s long-term memory or intermediate data, thereby altering its subsequent behavior based on externally plausible but false information [2]. Another critical aspect is insecure inter-agent interaction. In the absence of proper authentication, encryption, and trust verification, inter-agent communication channels become vulnerable to interception, spoofing, or message injection. This may lead one agent to perform actions imposed by an attacker through a compromised channel. In multi-agent systems, such failures can propagate in a cascading manner:

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a malfunction or compromise of one component triggers a series of incorrect decisions in others, ultimately resulting in system-wide failure.

A separate category involves the exploitation of human trust in agents. Autonomous systems are often perceived as competent and authoritative sources of recommendations, creating conditions for the manipulation of user behavior. This risk is amplified by the agent's ability to engage in context-adaptive interaction and by the effect of anthropomorphization, which may cause users to uncritically accept incorrect or covertly harmful recommendations.

Finally, so-called "uncontrolled" agents pose a significant risk. These are agents that deviate from prescribed policies, objectives, or constraints and begin to act beyond their original instructions [3]. Such scenarios are particularly dangerous in multi-agent environments, where the lack of strict isolation allows one agent to influence other components, shared memory, tools, or external APIs. Collectively, these threats demonstrate that the security of agentic systems must account not only for classical software vulnerabilities but also for the specific characteristics of autonomous planning, tool use, context exchange, and inter-agent interaction.

RAG-based systems introduce a distinct vulnerability surface because model behavior depends on externally retrieved content: poisoned documents can manipulate context and persist across sessions, sensitive data can be exposed through semantic retrieval and weak access controls, upstream data and embedding pipelines create supply-chain risk, and iterative probing can reconstruct indexed secrets through query-based extraction. These issues span prompt injection/context poisoning, data exfiltration, supply-chain compromise, and abuse of agent capabilities, so RAG security should treat retrieved content as untrusted input and enforce provenance tracking, access control, sanitization, scope limits, and anomaly monitoring.

The vulnerabilities outlined in the OWASP framework are presented in Tab. 1.

**Table 1**

| Attack Type | Target | Attack Vector | Protection |
|---|---|---|---|
| ASI01 Goal Hijack | Agent goals and priorities | Indirect prompt injection via web, docs, or tool outputs | Independent policy gate, explicit approval, drift logging |
| ASI02 Tool Misuse | Tool chains and integrations | Malicious parameters, tool cascades, supply chain compromise | Tool allowlists, blast radius limits, egress restrictions |
| ASI03 Identity Abuse | Tokens and session contexts | Token reuse, confused deputy, transitive privilege creep | JIT & short-lived tokens, task-scoped perms, re-validation |

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| Attack Type | Target | Attack Vector | Protection |
|---|---|---|---|
| ASI04 Supply Chain | Dependencies and registries | Descriptor spoofing, typosquatting, compromised packages | Version/hash pinning, artifact signing, kill switches |
| ASI05 RCE | Code execution environments | Unsafe eval, auto-run, filter bypass via encoding | Strict sandboxing, network isolation, Gen/Exec separation |
| ASI06 Memory Poisoning | RAG and long-term memory | "Poisoning" retrieval with false long-term patterns | Tenant segmentation, trust scoring, TTL/expiry, rollback |
| ASI07 Inter-Agent Comm | Protocols and mutual trust | Spoofing, replay attacks, descriptor tampering | mTLS/PKI, message signing, anti-replay (nonces) |
| ASI08 Cascading Failures | System-wide resilience | Error propagation through shared plans and memory | Circuit breakers, Planner/Executor split, lineage logs |
| ASI09 Trust Exploitation | Human approval process | Social engineering, "persuasive" lies, consent laundering | Approval with diff/plan view, multi-step verification |
| ASI10 Rogue Agents | Long-term behavior control | Hidden goal drift, agent collusion, slow escalation | Behavioral baselines, platform kill switch, goal audits |

A critical security risk in tool-augmented AI agents arises when three conditions coincide: access to sensitive user data, exposure to untrusted external content, and the capability to communicate with external systems [blog]. In such configurations, maliciously crafted inputs may influence the model's behavior and induce unintended disclosure of confidential information. This vulnerability is especially relevant in agentic environments that aggregate heterogeneous tools and data sources, because the model cannot always reliably distinguish operator instructions from adversarial content embedded in emails, web pages, documents, or other inputs. As a result, prompt injection becomes a practical exfiltration vector rather than a purely theoretical weakness, and conventional guardrails may be insufficient to guarantee robust protection.

To prevent vulnerabilities and attacks on AI agents as outlined by OWASP, it is advisable to implement a multi-layered system of controls covering access management, execution, data exchange, and monitoring. First and foremost, the principle of least privilege must be enforced: tool permissions should be restricted,

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short-lived tokens should be used, critical actions should require re-authorization, and high-risk operations must involve human confirmation. System prompts and agent goals should be strictly controlled, and any deviations from the original intent must be verified, logged, and, if necessary, blocked. Additionally, session and memory isolation between tasks should be ensured, along with centralized access policies and binding of tokens and permissions to a specific subject, resource, purpose, and time.

To mitigate risks associated with tool execution and external data, it is essential to employ sandboxing, policy-as-code enforcement, semantic validation of tool calls, input sanitization, resource usage controls, and comprehensive logging. Within the agentic supply chain, particular importance lies in component signing and attestation, the use of SBOM/AIBOM, allowlist-based dependency control, integrity verification of packages, and rejection of unsigned or untrusted artifacts. For inter-agent communication, it is recommended to use PKI, mTLS, message signing, and validation of nonces, timestamps, and session identifiers to prevent replay attacks, as well as gateway-based control of versions and capabilities.

Special attention should be given to protection against context poisoning and cascading failures. This includes data provenance tracking, anomaly detection, versioning, rollback mechanisms, separation of viewing and execution modes, and verification of results before further actions are taken. Effective measures also include adopting a Zero Trust approach, isolating agents, enforcing Just-in-Time access, applying rate limiting and circuit breakers, continuous monitoring, and maintaining a kill switch for the immediate disconnection of compromised components. Collectively, these measures provide a practical foundation for reducing risks related to unsafe behavior, privilege escalation, context compromise, and uncontrolled autonomy in AI agent systems.

Current approaches to securing agentic AI systems indicate a shift toward integrating model capabilities with mechanisms for controlled access, staged deployment, and operational risk containment. In particular, the Anthropic report notes that the experimental system Claude Mythos [12] is capable of autonomously identifying and exploiting zero-day vulnerabilities in widely used software platforms, including operating systems and web browsers, as well as generating functional exploits following an initial task specification. At the same time, it is emphasized that such research is conducted under conditions of strict execution isolation (containerized environments), employing multi-layered agentic architectures (agentic scaffolding), along with result verification procedures performed by separate agents and subsequent coordinated vulnerability disclosure in accordance with the principles of responsible disclosure. Initial system access is intentionally limited, being restricted through specialized programs targeted at vetted partners and developers, which aligns with a risk minimization strategy aimed at preventing premature or uncontrolled dissemination of high-risk capabilities.

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Similarly, OpenAI's approach [13] to scaling access for cybersecurity tasks reflects the institutionalization of the "trusted access" model as a key security mechanism. The Trusted Access for Cyber program provides the provision of enhanced capabilities of specialized models, such as GPT-5.4-Cyber, exclusively to verified entities, including individual researchers and organizations operating in the field of cybersecurity. In this context, a reduction in refusal boundaries for legitimate use cases is emphasized, particularly in areas such as binary analysis and reverse engineering, which are critical for defensive applications. At the same time, access to such capabilities is implemented through a multi-layered system of verification and gradual deployment, enabling a balance between increasing the effectiveness of defensive tools and mitigating the risks of potential misuse.

Google's approach to AI agent security [14] formalizes a structured framework aimed at constraining the operational risks of increasingly autonomous systems while preserving functional utility. The proposed model is organized around three core security principles: ensuring explicit human control over agent behavior, enforcing strict limitations on agent capabilities relative to task context and user-defined risk tolerance, and enabling end-to-end observability of both actions and intermediate decision processes for auditing and accountability. These principles are operationalized through a layered defense strategy in which deterministic runtime policy enforcement serves as the primary control mechanism for governing tool use and high-risk actions, including contextual restrictions based on data provenance and recent agent activity. In parallel, the framework incorporates reasoning-based safeguards that leverage auxiliary model components to detect potentially unsafe inputs or behaviors, particularly in cases involving novel or ambiguous attack patterns such as prompt injection. However, it is explicitly acknowledged that such probabilistic mechanisms cannot provide formal guarantees and therefore must function as complementary layers rather than standalone defenses. Overall, the approach reflects a systematic attempt to balance autonomy and security through explicit control boundaries, dynamic permission scoping, and increased transparency in agent execution.

Modern research in artificial intelligence security demonstrates a paradigm shift in evaluating system robustness toward resource-oriented models. An independent analysis of Claude Mythos conducted by the UK AI Safety Institute confirms the high effectiveness of such models in identifying security vulnerabilities [15]. A key empirical observation is the direct relationship between result quality and the volume of computational resources (tokens) expended, which creates a strong economic incentive to increase investment in security auditing processes. In this context, ensuring system security can be interpreted as a cost-optimization problem: it is necessary to invest more resources in discovering potential exploits than attackers are likely to spend on exploiting them in practice. This logic brings cybersecurity

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closer to the concept of Proof of Work, where the level of security is directly determined by the amount of "work" performed. An additional consequence is the increasing value of open-source software, as the costs of auditing and hardening it can be distributed across a wide user base, thereby improving resource efficiency and challenging the assumption that the reduced cost of developing alternatives diminishes the relevance of open-source projects.

**Conclusions.** The transition toward generative AI and agentic architectures fundamentally reshapes the cybersecurity threat landscape by expanding attack surfaces beyond traditional software vulnerabilities to include human interaction channels, autonomous decision-making processes, and complex multi-component ecosystems. The convergence of shadow AI usage, indirect prompt injection, tool misuse, supply chain compromise, and inter-agent communication risks demonstrates that modern threats are systemic rather than isolated, often propagating across organizational and technical boundaries. As illustrated by emerging incidents and OWASP-defined vulnerability classes, security failures in agentic systems stem not only from implementation flaws but also from inherent architectural properties such as autonomy, contextual memory, and external tool integration. Addressing these challenges requires a paradigm shift from conventional perimeter-based protection toward holistic, continuously adaptive security frameworks that incorporate strict access control, provenance verification, runtime monitoring, and robust isolation mechanisms across all layers of AI-driven infrastructures.

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*Дата першого надходження статті до видання: 11.04.2026*
*Дата прийняття статті до друку після рецензування: 26.04.2026*